
Glass-TA: Model benchmark for polymer glass transition temperature prediction

Arthur Benard, Arnau Casalprim, Simon Sermet-Magdelain and Matéo Tatzber

Abstract

In this work, five machine learning models were benchmarked: Ridge regression, Elastic Net, Random Forest, Support Vector Machine, and eXtreme Gradient Boosting, on a curated dataset of 7363 polymers from the PolyMetriX database. Each polymer is featured as a 2048-bit RDKit Morgan fingerprint computed from its PSMILES string, and models are optimized by grid search on a held-out validation set before final evaluation on a test set. Random Forest achieved the best performance (RMSE = 41.89 K, MAE = 27.03 K), outperforming all other approaches. Linear models with regularization provided competitive baselines (RMSE $\simeq$ 44 K), while SVM and XGBoost showed higher errors. All models exhibit increased prediction uncertainty above 400 K, attributable to limited training data in that range. These findings suggest that Morgan fingerprints capture meaningful structure–property relationships for T_g prediction, while pointing to the need for richer polymer representations and improved high- T_g data coverage in future work.

1 Introduction

Nowadays, polymers take a very important place in the world we live in, they are large molecules that are made of repeating structural units. Their properties vary depending on their chemical structures and resulting physical properties but they are present on a daily basis in all kinds of material, showing their indisputable versatility and their indispensability. By modifying their chemical structures (the monomer pattern), they can be engineered for almost any application, going from aerospace parts to flexible electronics. One of the most important properties we need to know when designing a new polymer is the glass transition temperature characteristic of each polymer. This property is important because it will determine the polymer's potential use depending on the temperature application.

The glass transition temperature (T_g) of a polymer is the temperature at which a solid amorphous polymer transitions into a flexible, rubbery state. At this T_g , the polymer chain starts to rotate and be more flexible. Knowing T_g is therefore essential for ensuring that a material performs correctly under its operating conditions.

However, measuring T_g experimentally is both costly and time-consuming, and no simple analytical formula directly links a polymer's chemical structure to its T_g . This makes machine learning an attractive approach: given a large enough dataset of known polymers, a model can learn the relationship between molecular structure and thermal behavior, enabling fast and inexpensive predictions for new candidates.

The main objective of our project is to predict T_g to guide the selection of polymers for applications with specific thermal requirements. In order to predict the glass transition temperature, a first try was performed with a standard linear regression (Ordinary Least Squares) and then five different machine learning models were implemented: Ridge regression, Elastic Net, Random forest, Support Vector Machine and eXtreme Gradient Boosting. Each model will be evaluated using RMSE and MAE, and the best-performing approach will be identified and discussed.

2 Methods

A curated dataset of 7367 polymers represented as polymer smiles (PSMILES) along with their corresponding glass transition temperature was obtained from PolyMetriX¹ paper. Before training the models, the dataset has been filtered to remove the data considered as unreliable (labeled red) resulting in a final dataset of 7363 polymers. Then, the filtered data was separated into three sets: the training set (70%), the validation set (15%) and the test set (15%). The performance of different models will be assessed by tuning and optimizing their hyperparameters. Each model is

trained using a combination of hyperparameters on the training dataset, and the best set of hyperparameters is selected based on the root mean square error of the prediction on the validation dataset. Then, the best model is retrained using the concatenated training and validation sets. The final model is saved as a joblib file, and its performance is evaluated by predicting on the test set.

The PSMILES were featurized as Morgan fingerprint² vectors (2048 bits and a radius of 2) following the general idea proposed by C. Kuenneth et al.³ allowing one to transform a PSMILES into a mathematical object. This workflow was reproduced using the 5 different models (there are actually 6 models proposed but linear regression in closed form has no hyperparameters and results in a singular matrix) presented below.

2.1 Linear and Ridge regression

Initially, we evaluated a standard linear regression⁴ (Ordinary Least Squares) in closed form, this model calculates a linear combination of all 2048 features, finding the precise set of weights that minimizes the sum of squared residuals between the predicted temperatures and the true experimental values. However, because our data set contains 2048 different chemical features, this model has a tendency toward multicollinearity and overfitting resulting in an unsolvable linear system. Multicollinearity occurs because many of the fingerprint features represent overlapping molecular substructures, creating an unsolvable linear system where the model struggles to isolate the true chemical causes of thermal changes.

To solve this, the Ridge regression model was implemented from scratch in its closed form. This model adds an L_2 regularization penalty that will force the algorithm to keep its learned weights as small as possible, preventing redundant chemical features from destabilizing the math.⁴

$$\text{Loss} = \text{MSE} + L_2 = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \alpha \sum_{j=1}^p w_j^2 \quad (1)$$

Where Loss is the total error the model is trying to minimize during training, MSE is the mean squared error, L_2 is the regularization penalty, n is the number of training samples, y_i and $\hat{y}_i$ are respectively the true and predicted glass temperature for polymer i , α is the regularization hyperparameter that controls how strict the penalty is, p is the number of features (2048 in our case) and w_j is the weight of feature j .

2.2 Elastic net

Elastic net⁵ is a regularized regression model that combines the strengths of both Lasso (L_1) and Ridge (L_2) penalties. As explained earlier, Ridge penalty (L_2) handles multicollinearity by shrinking weights while Lasso penalty (L_1) addresses the Morgan fingerprint by ignoring irrelevant features entirely. Consequently, by applying both penalties, Elastic Net successfully eliminates useless molecular structures while maintaining mathematical stability among groups of correlated features.

$$\text{Loss} = \frac{1}{2n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \alpha \cdot l_1 \sum_{j=1}^p |w_j| + 0.5\alpha \cdot (1 - l_1) \sum_{j=1}^p w_j^2 \quad (2)$$

Where α is the hyperparameter that controls the overall strength of the combined regularization penalty and where l_1 is the hyperparameter that balances the mix between the two penalties.

2.3 Random forest

Random forest model⁶ is an ensemble learning algorithm that focuses on the construction of independent decision trees during training. A single decision tree works by continuously splitting the PolyMetriX¹ dataset into smaller groups based on the presence or absence of specific chemical features. At each step, the tree mathematically selects the specific feature that results in the greatest reduction in variance between the original data and the newly split subsets. To reduce correlation between trees, random forests use bootstrap aggregating, or bagging. In the scikit-learn implementation used here, each tree was trained on a bootstrap sample of the training polymers. Since the default value `max_features=1.0` was kept, all Morgan fingerprint features were considered when selecting each split. Therefore, the diversity between trees mainly arises from the different bootstrap samples used during training. The final predicted temperature of the random forest model is obtained by averaging the predictions of all individual trees.

Several hyperparameters characterize this model and three of them were tuned during training: the `n_estimators` parameter, which defines the total number of trees in the forest, and `max_features`, which corresponds to the total number

of features the trees are allowed to choose from during each branching step. Regarding the depth of each tree, the *max_depth* hyper, parameter fixes the maximum number of consecutive splits a tree is allowed to make. Finally, *min_samples_split* and *min_samples_leaf* dictate the minimum number of polymers required in a node to authorize a split, and the minimum number of polymers required to form a terminal leaf, respectively. In order to reduce computational cost, only *n_estimators*, *max_depth* and *min_samples_leaf* were tuned during validation.

2.4 Support Vector Machine (SVM)

An SVM⁷ works by finding the optimal boundary (called hyperplane) that maximizes the distance between different classes of data. To ensure the cleanest split, it relies entirely on the closest data points to that boundary, known as support vectors. When data is too chaotic to separate with a linear fit, it uses a mathematical shortcut to map the points into a higher dimension where they can be cleanly divided. The width of the tolerance margin and the penalty for violations are controlled by two key hyper parameters: ϵ that defines the insensitivity tube around the regression line, and C which balances model complexity against training error, a large C forces the model to fit the training points more closely at the risk of overfitting.

2.5 eXtreme Gradient Boosting

XGBoost⁸ is an optimized implementation of gradient boosting, where an ensemble of decision trees is built sequentially. Each new tree is trained to correct the remaining prediction error of the previous ensemble. The final prediction is obtained as the sum of the contributions of all trees. The key hyperparameters tuned were the number of trees (*n_estimators*) and the learning rate (which scales each tree’s contribution to make the boosting process more conservative). The maximum tree depth (*max_depth*, which controls model complexity) was set to 3 for all trainings.

3 Results & Discussion

To evaluate the models, RMSE and MAE were calculated for each of them and a comparison was performed. With the mentioned hyperparameters to tune in methods section, models have been assessed and the RMSE for the validation set was plotted in Figure 3.

The performance of the five best models evaluated before is summarized in Table 1:

Table 1: Comparison of all models evaluated on the test set. Optimization and fitting were executed using the training and validation subsets while reported evaluation metrics are from prediction of the trained model on test dataset.

Model	Hyper parameters	RMSE [K]	MAE [K]
Ridge regression	$\alpha = 10$	44.17	31.38
Elastic Net regression	$\alpha = 0.001$ $\ell_1 = 0.1$	44.49	31.45
Random Forest	<i>max_depth</i> = 40 <i>n_estimators</i> = 500 <i>min_samples_leaf</i> = 1	41.89	27.03
SVM	$C = 1000$ $\epsilon = 0.5$	48.35	31.05
XGBoost	<i>max_depth</i> = 3 <i>n_estimators</i> = 300 <i>learning_rate</i> = 0.1	46.23	32.93

Comparing these models, one of them stood out among the others: the RF had the lowest RMSE and the lowest MAE. This result seems normal: Morgan Fingerprints are a large 2048 bit vector representing the presence of an aromatic group or not, for example. Each tree can select the structure element that most impacts the T_g even for non-linear behavior. Instead, the Ridge and Elastic Net models have quite comparable results, and the linear approximation seems to be a quite good first approximation. Considering XGBoost, its RMSE and MAE are high compared to the others, which is quite unpredictable because this type of model is often better than RF one. This may be due to small max depth of only 3 compared to much deeper trees for RF (40).

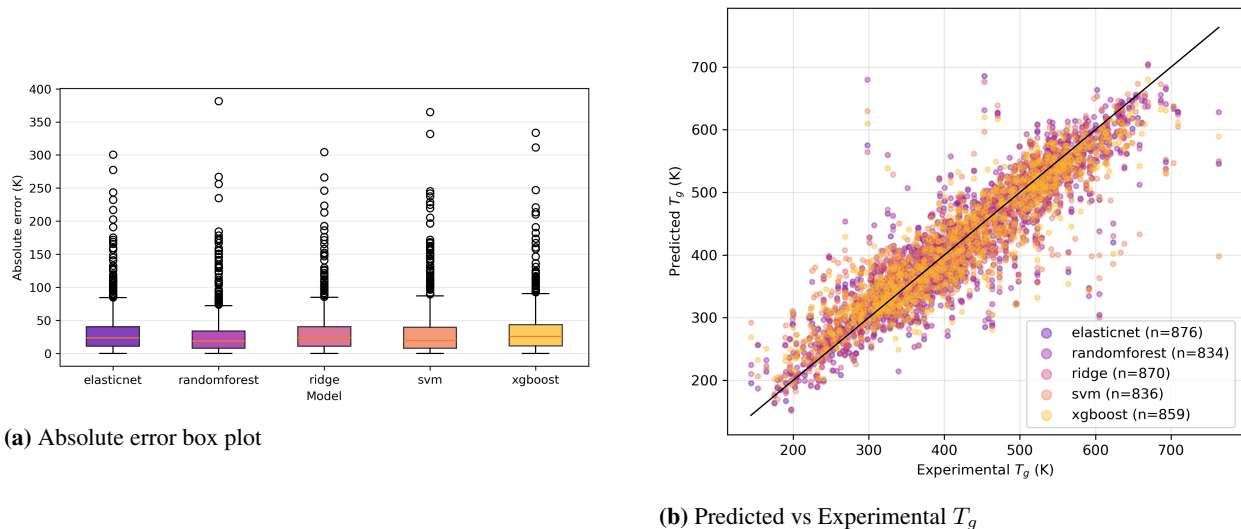


Figure 1: Test performance and error distribution across the models.

The models can also be compared with their outlier predictions. Considering Figure 1a, each model has a similar interquartile range of absolute error, but, for example, RF, SVM and XGBoost have larger outliers with a deviation over 300 K (absolute error) for a few polymers. Then, in Figure 1b, one can see that experimental values have lower prediction error below approximately 400 K. Above this threshold, there are more outliers for all models, which often underestimate the experimental values. The lack of precision above this threshold is likely due to the lack of data available at high T_g (around 450 K and above 550 K, in Figure 2), limiting the regression at high T. A second potential improvement is to not only use Morgan Fingerprints to predict the T_g values: Morgan fingerprints only encode local atomic environments and can capture the presence of specific substructures, such as aromatic rings or halogen-containing environments². However, they do not highlight important polymer properties such as tacticity or molecular weight distribution. Thus, models’ performance could be improved by enlarging the dataset and exploring new representations such as different fingerprint types or fixed-length token-based embeddings.

4 Conclusion

This work benchmarked six machine learning models for predicting the glass transition temperature of polymers from Morgan fingerprint representations of their PSMILES strings. Standard linear regression was immediately ruled out due to the multicollinearity inherent in the 2048-bit fingerprint space, confirming the need for regularization. Ridge and Elastic Net regression produced comparable results ($\text{RMSE} \simeq 44$ K), demonstrating that a simple linear baseline with regularization already captures a significant share of the structure–property relationship. Among the more complex models, Random Forest achieved the best performance overall with an RMSE of 41.89 K and an MAE of 27.03 K, benefiting from its ability to model non-linear interactions between chemical substructures. Notably, XGBoost underperformed relative to expectations, likely due to the constrained tree depth used during hyperparameter search. All models shared a common limitation: prediction accuracy degrades above approximately 400 K, where training data is sparse, causing systematic underestimation of high- T_g polymers.

These results highlight both the promise and the limits of Morgan fingerprints as a polymer representation. While they efficiently encode local chemical environments, they carry no information about tacticity, chain length, or molecular weight distribution, all of which are known to influence T_g .⁹ Future work could address this by incorporating descriptor-based features or graph-based representations, and by augmenting the dataset in the high- T_g regime where all current models struggle most.

5 Attachments

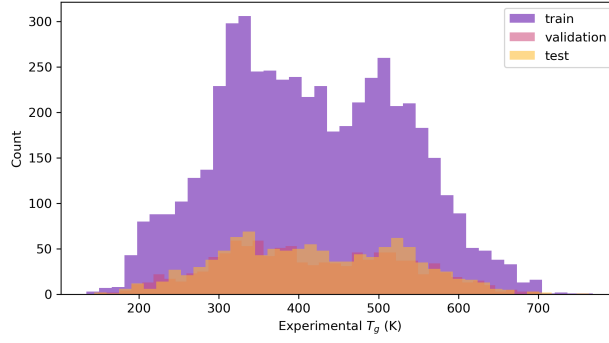
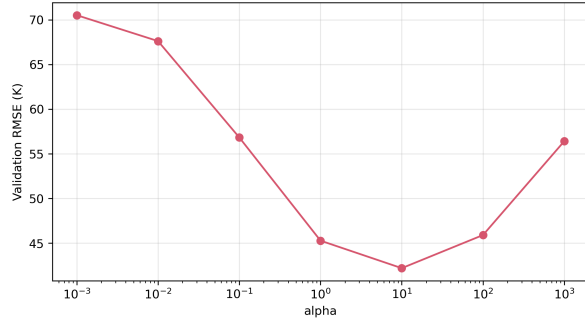
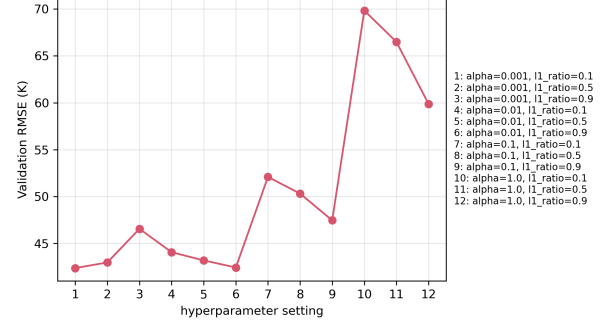


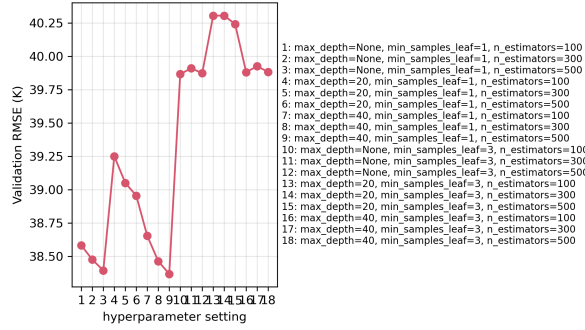
Figure 2: T_g distribution in the dataset



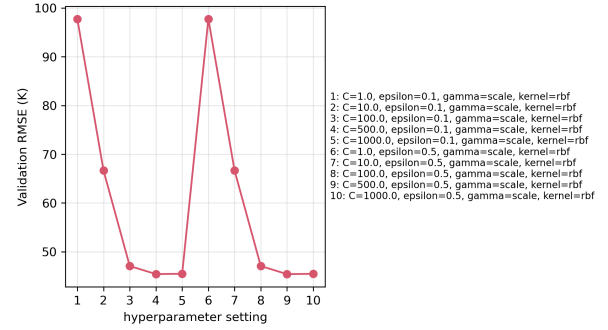
(a) Ridge regression



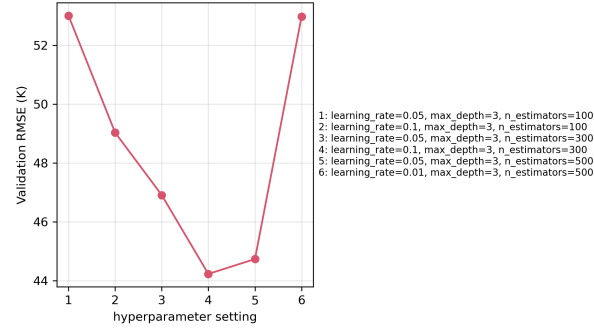
(b) Elastic Net



(c) Random Forest



(d) SVM



(e) XGBoost

Figure 3: Tuning of hyperparameters for model training on validation set.

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